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PhysioTwin: Stress Detection from Wearable Signals

A Three Stage Pipeline For Digital Twin iOS Application

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Motivation & Background

- 1 in 3 adults experience chronic stress, linked to cardiovascular disease and burnout
- Wearables report stress as a score, not actionable or real-time
- Exercise mimics stress signals (elevated HR, EDA, movement) — naive models fail
- Gap:** No system distinguishes cognitive stress from physical exertion in real-time

Problem Statement

- Goal:** Detect and differentiate between cognitive stress and physical stress from Apple Watch signals
- Challenge:** Feature gap between lab wearables (Empatica E4) and consumer devices
- Innovation:** 3-stage hybrid pipeline — ML + domain rules + validated formulas

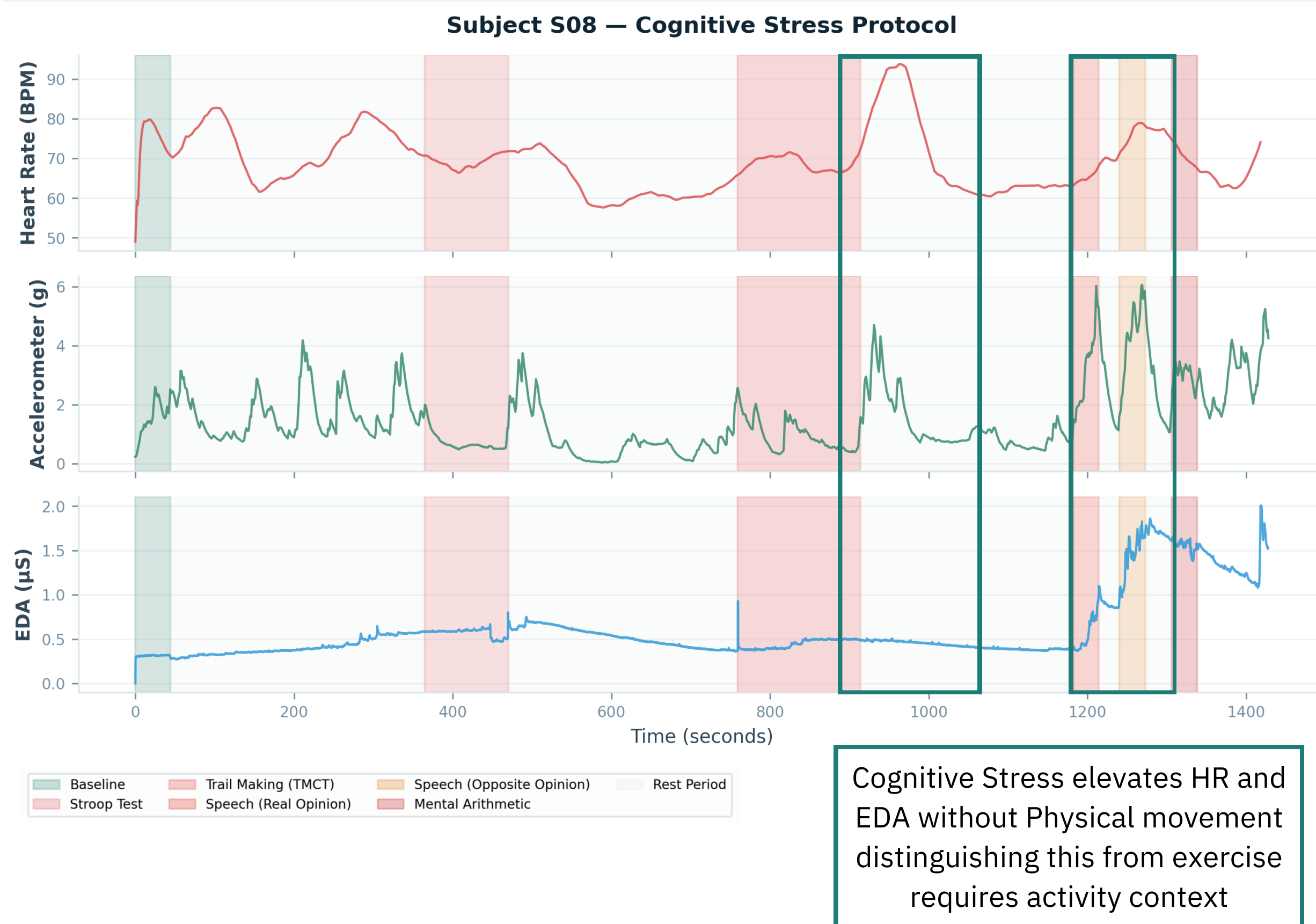
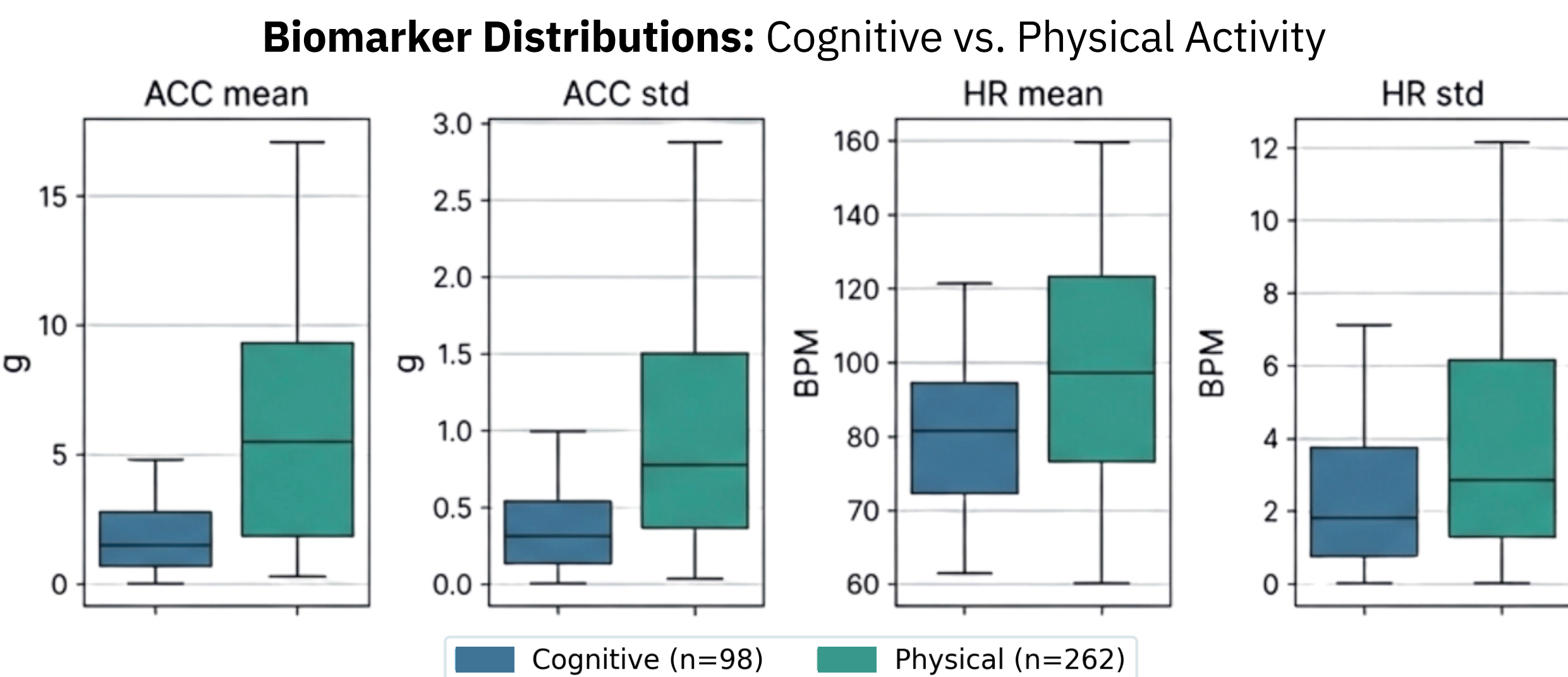


Figure: Raw biosignals during cognitive stress tasks (S08)

Dataset

Dataset	N Device	Use
WISE	22Empatica E4	Training(labeled)
Apple Health 1	Apple Watch S7	Deployment test

- WISE: 2,449 windows (30s, 50% overlap) from 3 protocols (cognitive stress, aerobic, anaerobic)
- Signals: HR, ACC, EDA, HRV (IBI), TEMP
- 2-class split: 413 COGNITIVE vs 2,036 PHYSICAL



3-Stage Pipeline Architecture

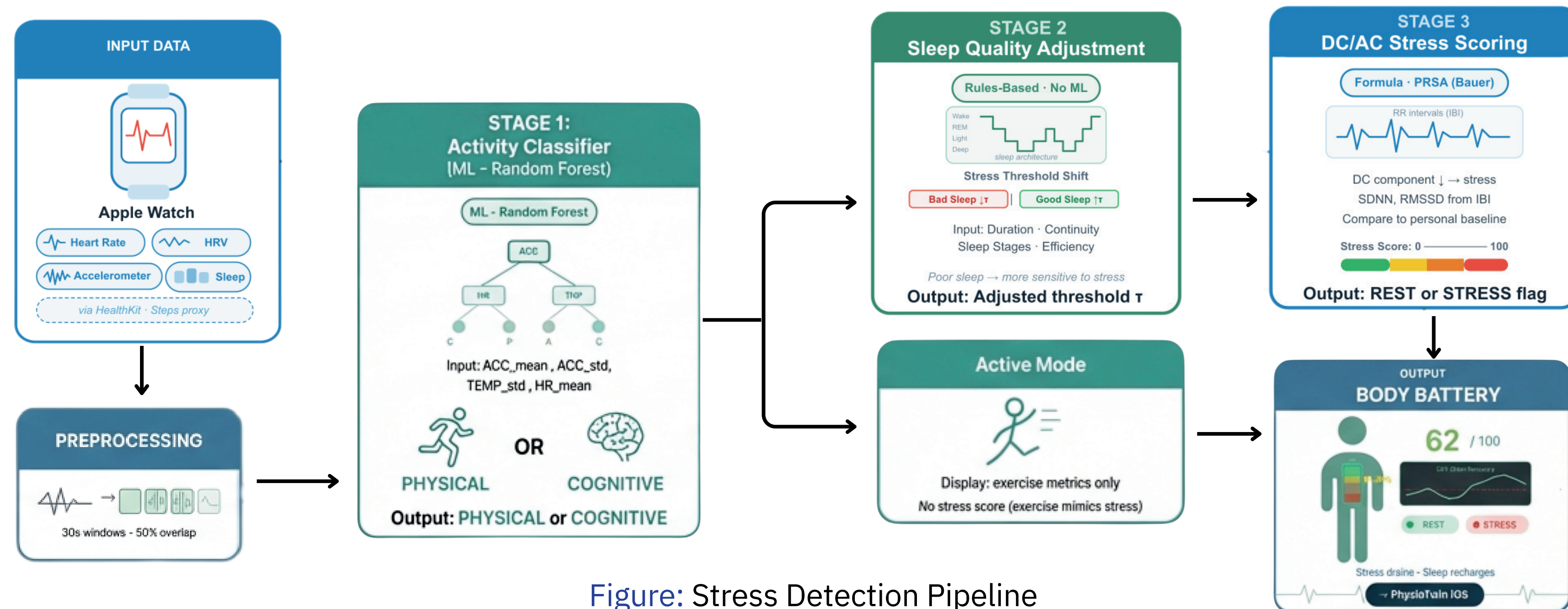
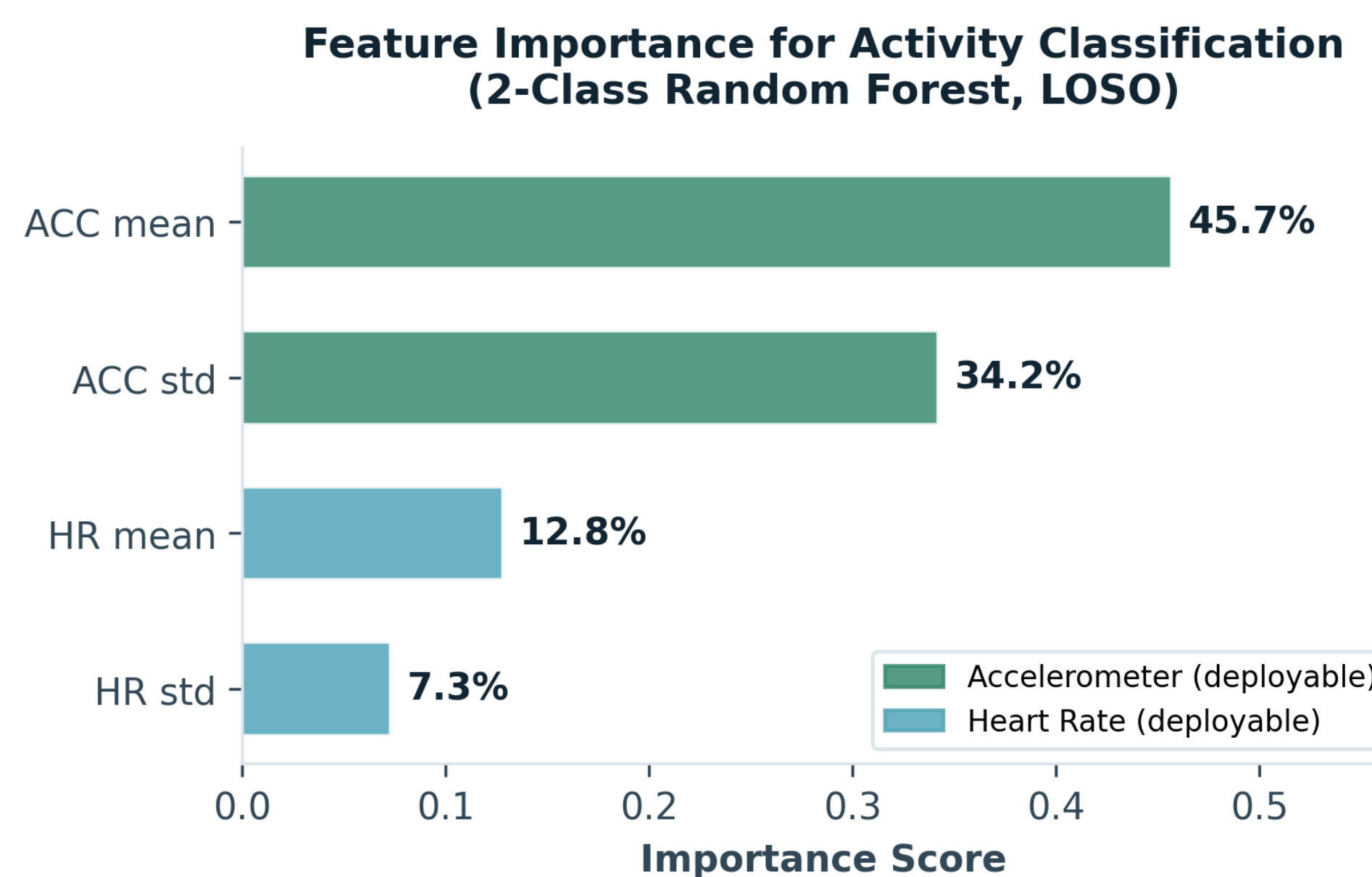


Figure: Stress Detection Pipeline

Stage 1: Activity Classifier ML - Random Forest

- Random Forest, 100 trees
- 30s sliding windows, 50% overlap
- Leave-One-Subject-Out validation (22 folds)
- Classifies: **PHYSICAL** vs **COGNITIVE**
- ACC (movement) is primary differentiator, not HR

Feature Set	LOSO Acc.
Full Sensors (10)	94.4%
Apple Watch (6)	94.3%
HR + ACC (4) [deployed]	93.6%
HR + HRV only (5)	81.5%



Stage 2: Sleep Quality Adjustment Rules Based

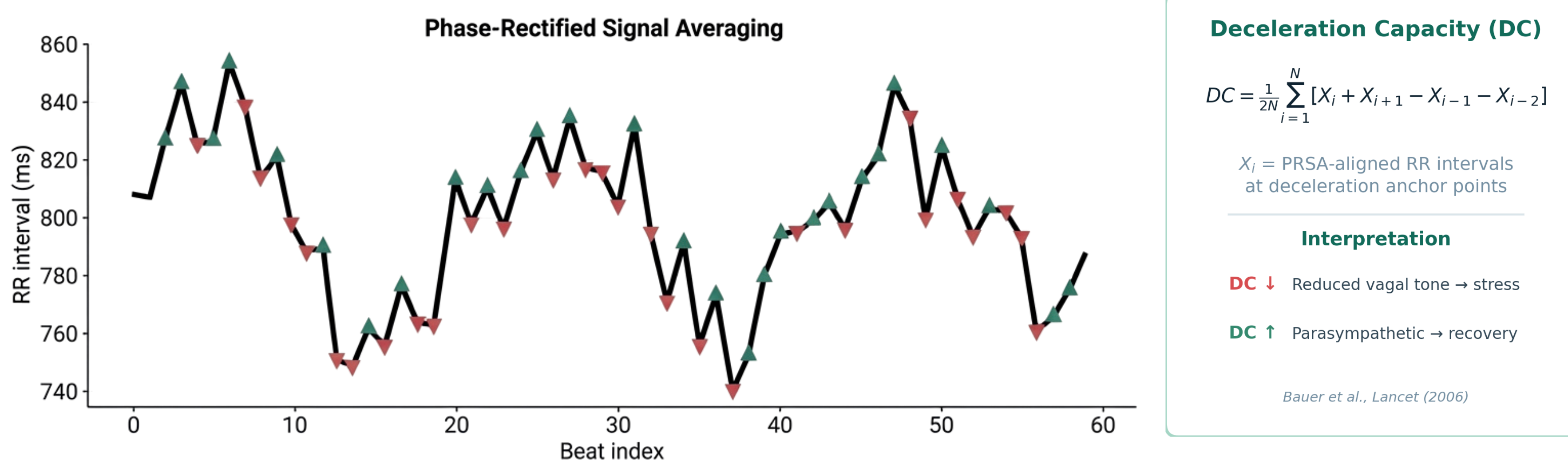
- We first compute sleep quality through metrics like quantity, continuity, deep/REM sleep and physio score.
- Through the sleep quality score, we recharge the battery first and then compare to personal 7-night baseline to adjust stress sensitivity
- The more poor a person's sleep is, the more sensitive they will be to stress, and we will drain their battery accordingly

Quality	vs Baseline	Multiplier	Effect
Very Poor	< 75%	×0.80	Much more sensitive
Poor	< 90%	×0.85	More sensitive
Normal	90–110%	×1.00	No change
Good	> 110%	×1.05	Less sensitive
Excellent	> 120%	×1.08	Less sensitive

Example: stress score 55 → flagged as **STRESSED** after 4.5h sleep (threshold 48), but **OK** after 9h sleep (threshold 64)

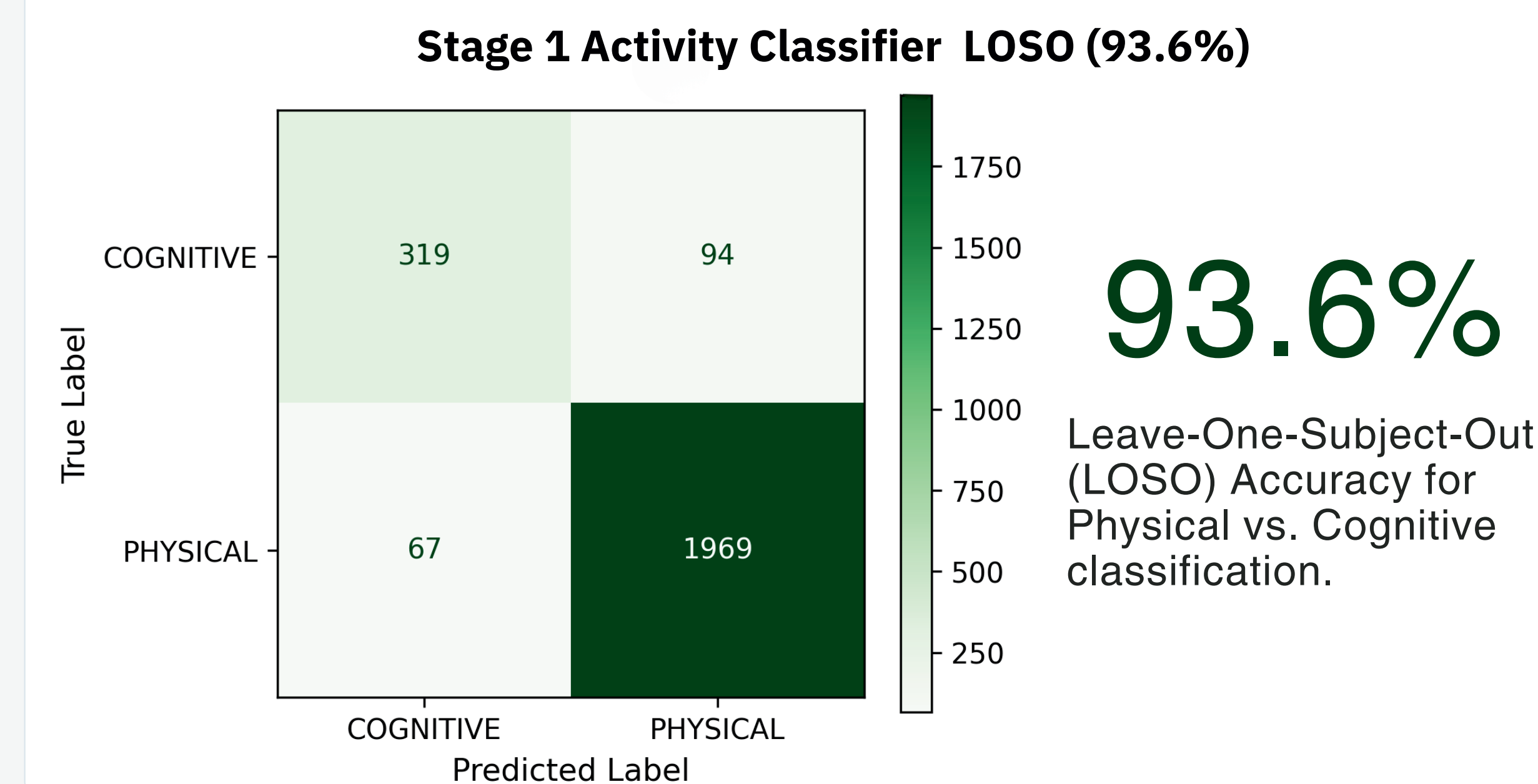
Stage 3: DC/AC Stress Detection Formula-Based

- Deceleration Capacity (DC) quantifies HRV asymmetry via Phase-Rectified Signal Averaging (PRSA)
- Stress reduces parasympathetic activity → DC drops below sleep-adjusted threshold
- Opportunistic: full DC/AC when data available;HR-only fallback



Results

Component	Metric	Value
Activity Clf.	LOSO Accuracy	93.6%
Activity Clf.	F1 Macro	0.88
Sleep Rules	p (eff. → stress)	+0.042 (n.s.)
DC/AC Formula	p-value (WISE IBI)	0.075 (n.s.)



Body Battery Demo

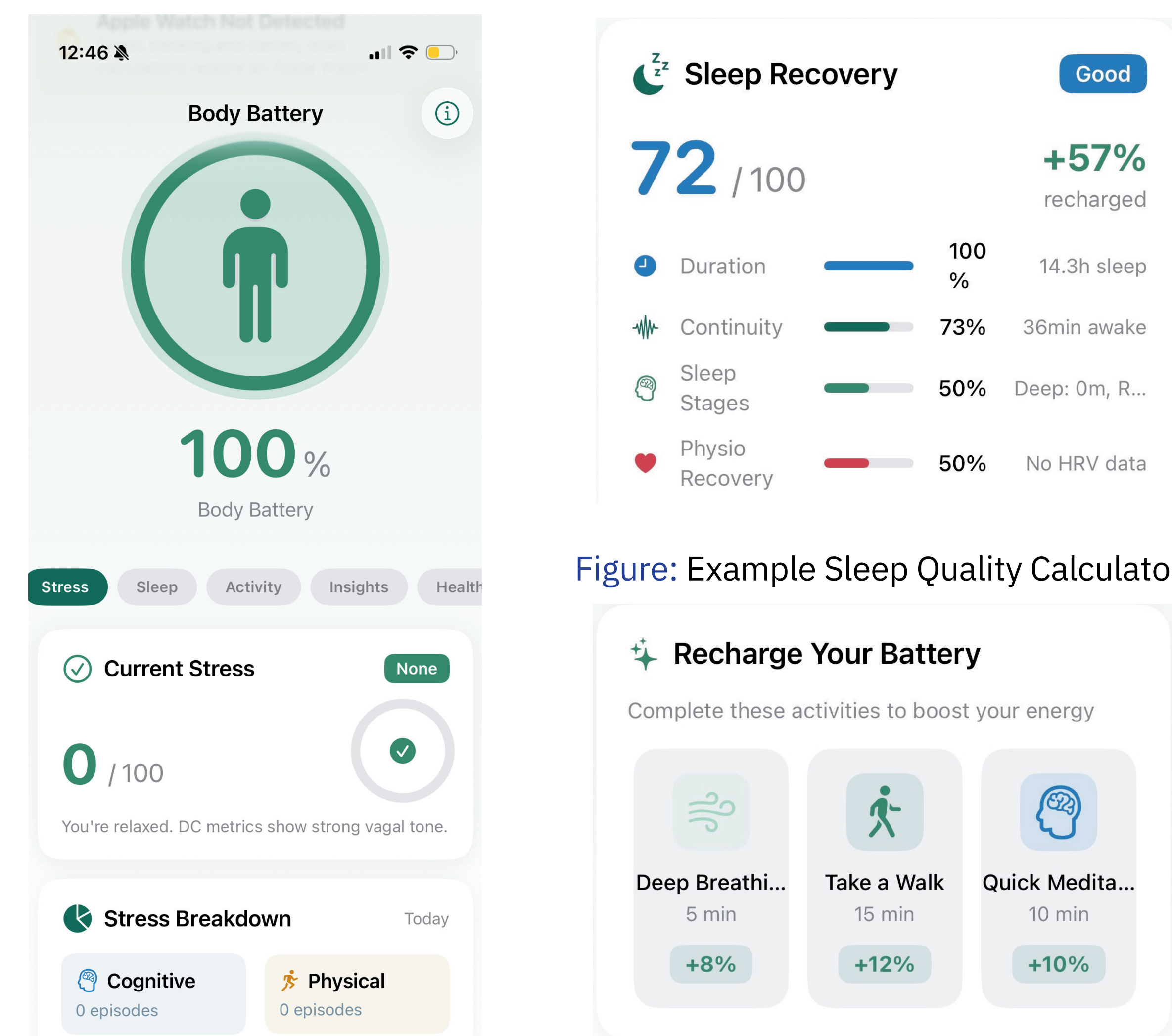


Figure: Home Page of iOS app

Figure: Activities to Recharge Battery

Conclusion & References

- PhysioTwin achieves 93.6% LOSO accuracy distinguishing cognitive vs. physical stress using only HR + ACC from consumer wearables, deployed on Apple Watch via an iOS app
- Personal baselines outperform population-level thresholds — both sleep adjustment and DC/AC results support
- individualized stress profiling over fixed cutoffs.

References:

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- [4] Schmidt, P. et al. (2018). WESAD: Multimodal dataset. ICMI